

WAM CONTACT COUNTERFACTUALS: BRANCH-STRUCTURED EVIDENCE FOR EMBODIED WORLD-ACTION MODELS

Anonymous authors

Paper under double-blind review

ABSTRACT

We study world-action models for contact-rich manipulation. The motivating bet is: Treat unobserved contact alternatives as first-class latent futures, not data augmentation.. We introduce a mechanism-level representation that keeps physically admissible but unrealized action outcomes as explicit branches. A synthetic contact-style evaluation shows that the proposed branch mechanism improves hidden-mode success and tail-risk relative to observed-only, augmentation-as-data, and scalar-uncertainty baselines.

1 INTRODUCTION

Robotics papers often collapse unobserved physical alternatives into extra data, variance, or a single predicted next state. For world-action models for contact-rich manipulation, this can erase the difference between modes that are visually similar but action-critical. This paper asks whether a world-action model should expose those alternatives to the planner before execution.

2 HOSTILE PRIOR WORK

The closest hostile records include (pri, 2026a;b; 2025a;b; 2026c; 2024). They make generic world models, contact prediction, and uncertainty insufficient as novelty claims. Our boundary is narrower: preserve branch-valued contact futures as the planning object.

3 MECHANISM

Given state s and action a , the model returns an observed next-state prediction plus a finite atlas $B(s, a)$ of admissible latent branches. Planning minimizes nominal loss plus adverse-branch tail risk. This is not bigger data or a new benchmark; it changes the object represented by the WAM.

4 EVIDENCE

We include a runnable synthetic testbed in `src/run_experiment.py`. It compares four variants under nominal and hidden-mode-shift conditions.

Variant	Split	Success	Tail risk
observed_only_wam	nominal	0.590	0.410
observed_only_wam	hidden_mode_shift	0.170	0.830
augmentation_as_data_wam	nominal	0.570	0.430
augmentation_as_data_wam	hidden_mode_shift	0.250	0.750
scalar_uncertainty_wam	nominal	0.667	0.333
scalar_uncertainty_wam	hidden_mode_shift	0.333	0.667
proposed_branch_mechanism	nominal	0.770	0.230
proposed_branch_mechanism	hidden_mode_shift	0.588	0.412

5 LIMITATIONS

The evidence is synthetic. It supports the mechanism boundary and failure mode, not real-robot state of the art. Hardware validation, richer contact geometry, and learned branch discovery remain open.

6 CONCLUSION

WAM Contact Counterfactuals argues that embodied world-action models should preserve unrealized physical alternatives when those alternatives change downstream action value.

REFERENCES

- Planning-guided diffusion policy learning for generalizable contact-rich bimanual manipulation, 2024. <http://arxiv.org/abs/2412.02676v2>.
- Learning to feel the future: Dreamtacvla for contact-rich manipulation, 2025a. <http://arxiv.org/abs/2512.23864v3>.
- Forcevla: Enhancing vla models with a force-aware moe for contact-rich manipulation, 2025b. <http://arxiv.org/abs/2505.22159v3>.
- Data-efficient and contact-rich manipulation through diffusion augmentation and vision-language models, 2026a. <https://doi.org/10.1609/aaai.v40i47.41353>.
- Omnivta: Visuo-tactile world modeling for contact-rich robotic manipulation, 2026b. <http://arxiv.org/abs/2603.19201v2>.
- Dexvitac: Collecting human visuo-tactile-kinematic demonstrations for contact-rich dexterous manipulation, 2026c. <http://arxiv.org/abs/2603.17851v1>.